

8 Spatial Networks

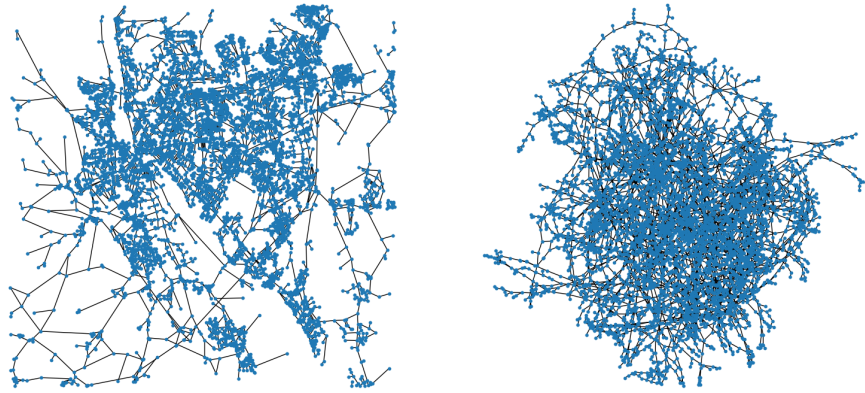


Figure 1: Left is the Exeter road network where positions are geographical. Right is the same network using a force directed layout. Exeter is built on either side on an estuary which can be seen in the left plot but vanishes in the hairball on the right.

Unlike many of the networks we have discussed so far like social or technological ones many networks are embedded in the real world. A concrete example is a road network like Figure 1. Such networks are qualitatively different from social networks because of **steric constraints** - for the road network you can't have an intersection where 400 roads meet, usually 4 is the maximum! Thus some of the metrics developed previously, which strongly emphasise degree, need to be swapped with others more suitable for these physical or spatial networks.

Reading

This material is covered in a lot more depth in [1].

8.1 Basic definitions

A spatial network is just a graph G consisting of an edge set E and a vertex set V as usual along with a set of positions $X = \{x_1, \dots, x_N\}$, where x_i is a

(usually two dimensional) vector.

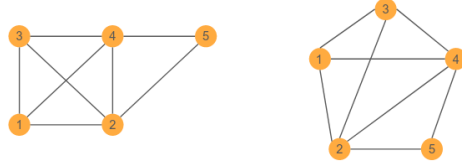


Figure 2: Same topology, different X .

Figure 2 is an example of two spatial networks with the same G but different X s. If only *topological* properties are relevant i.e. the pattern of connections, then they can be considered equivalent. But if spatial positions are important e.g. because the links represent something physical like roads or pipes, where length and position matter, then they are not equivalent.

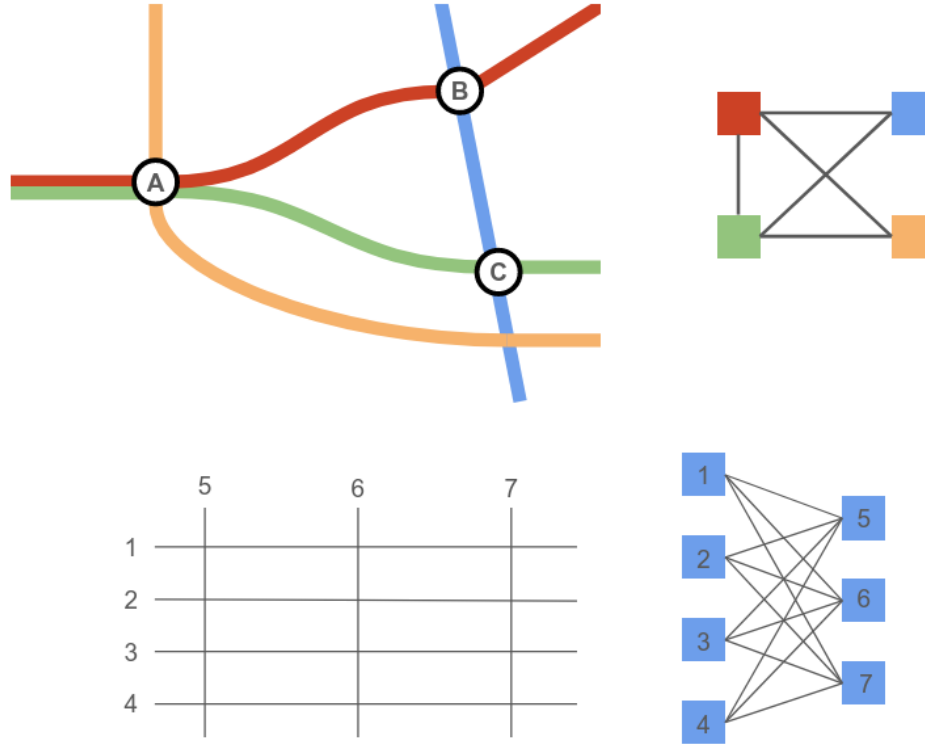


Figure 3: Top, a simplified metro map and its dual. Bottom, a simplified street map and its dual.

We may represent spatial networks directly, as in the map in Figure 1 or the simpler maps on the left of Figure 3 where the lines are the rails or roads and the nodes are train stations or street intersections. We can also use a more abstract representation, also shown in Figure 3, where the nodes are now the roads or rails and these are linked if they intersect at a node. This exchange of nodes to links and links to nodes is called the **dual graph**. We will see both are useful.

8.2 Planarity

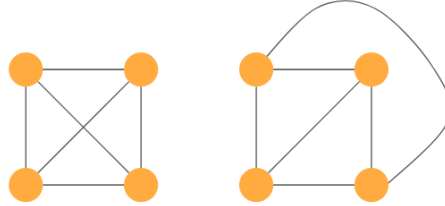


Figure 4: A non-planar and planar drawing of K_4 .

For now, forget about the spatial positions X for now and just study the connections. A graph is planar if it can be drawn in 2-d without any edges crossing each other. Knowing if a graph can have a planar, or nearly planar, drawing could be important for, say, routing connections on a computer chip or planning a bus route. Existing spatial networks are often planar, or nearly planar. Think of the road network, roads generally meet at intersections, overpasses and underpasses are rare and expensive to construct. Figure 4 shows that planar graphs can have non-planar drawings so we can't rely on the drawing alone to tell if a graph is planar. A famous result called **Kuratowski's Theorem** says that wherever the complete graph on 5 nodes, K_5 , or the complete bipartite graph with 3 nodes in each part, $K_{3,3}$, are subgraphs of G then G is non-planar, otherwise it is planar.

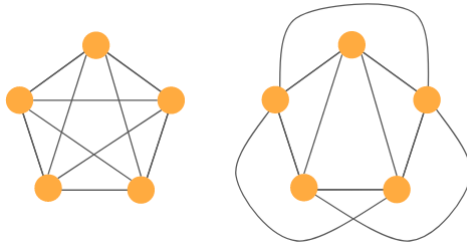
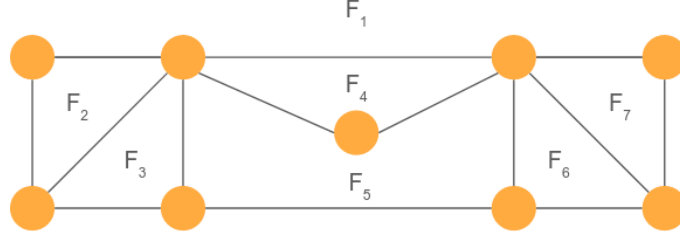


Figure 5: It is impossible to draw K_5 without at least 1 crossing.

The K_5 and $K_{3,3}$ graphs are the basic 'atoms' of non-planarity since they are the smallest non-planar graphs, see figure 5. Proving this is a standard result in graph theory textbooks, but a little involved so we won't do it here.

Figure 6: $N = 9, E = 16, F = 7$.

Planar networks satisfy **Euler's formula** which states

$$2 = N - E + F \quad (1)$$

Where $E \equiv L$ is the number of edges, N the number of nodes and F the number of **faces**, shown in Figure 6. Note we always include the 'outside face' i.e. the area outside the graph. Euler's theorem is also proved in standard introductions to graph theory. It is less tricky than Kuratowski's Theorem. The basic idea is that when you remove an edge N is fixed, $E \rightarrow E - 1$ and $F \rightarrow F - 1$ leaving the expression invariant. When you have N nodes and $N + 1$ edges left the formula follows.

A useful result derived from Euler's formula is a bound on the degrees of nodes in a planar graph. Because every face is enclosed by at least 3 edges we have

$$F \leq \frac{2E}{3}$$

Rearranging and substituting into Euler's formula

$$F = 2 - N + E \leq \frac{2E}{3} \rightarrow E \leq 3N - 6 \equiv \langle k \rangle \leq 6 - \frac{12}{N} \quad (2)$$

This tells us first, if a network has average degree greater than 6 it is not planar and second that planar graphs are sparse, the number of edges is not too much larger than the number of nodes.

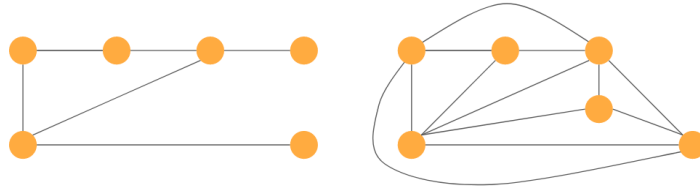
Figure 7: Left $N = 6, E = 6, F = 2$, Right $N = 6, E = 12, F = 8$.

Figure 7 shows a planar graph obtained from taking points and adding the maximum possible number of edges without making it non-planar. All the faces must be triangles (if not we could add extra edges to make them so). In this case $F = \frac{2E}{3}$ and we have from Euler's formula $E = 3N - 6$ and $F = 2 - N + 3N - 6 = 2N - 4$. The **meshedness coefficient** is

$$\alpha = \frac{F - 1}{F_{max} - 1} = \frac{E - N + 1}{2N - 5} \quad (3)$$

and measures how far from this maximum a real network is. A related coefficient is

$$\gamma_P = \frac{E}{E_{max}} = \frac{E}{3N - 6} \quad (4)$$

Taking the large N limit neither of these variables contain much more information than the mean degree $\langle k \rangle = \frac{2E}{N}$. Since this is what we are used to we will continue to use that instead.

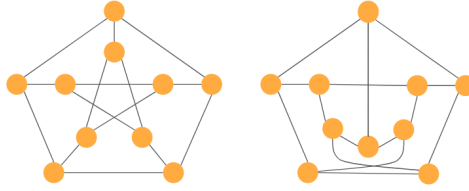


Figure 8: The Petersen graph.

For non-planar graphs we want to measure how far from planarity they are. This is a very practical problem, imagine we are organising wires on a chip, we have to avoid having loops than have to go other wires up into the air! For road and rail crossings and bridges are expensive. Figure 8 shows the famous Petersen graph drawn in two different ways, one has 5 crossings and the other has 2 crossings. For a spatial network, if the positions are fixed e.g. cities connected by rails, then we can just count the number of crossings. However if we have freedom to move the nodes then we can draw the network in such a way as to minimise crossings. The **crossing number** $cr(G)$ is the lowest number of edge crossings in a drawing of G . This number is very hard to compute.

Other measures of deviation from planarity are:

- **thickness**, the minimum number of planar subgraphs into which the graph can be decomposed

- **skewness**, the number of edges that one has to remove from a graph G to obtain the maximum planar subgraph G
- **splitting number** the smallest number k such that G can be obtained from planar graph G' by k vertex identifications.

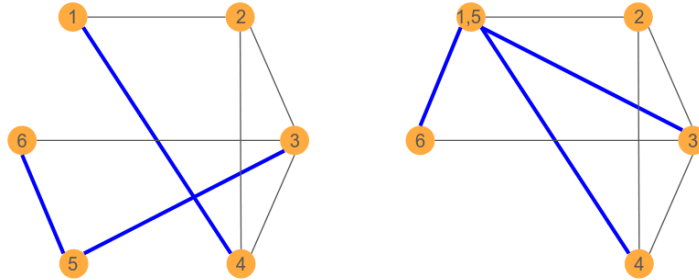


Figure 9: Contracting or identifying nodes 1 and 5 means deleting them and their edges, then making a new node 1,5 with all the edges that 1 and 5 had.

Figure 9 shows the meaning of vertex identification. For a planar graph the thickness, skewness and splitting number are 1, 0 and 0 respectively. In general however, all of these quantities are about as hard to compute as the crossing number.

8.3 One-way Streets

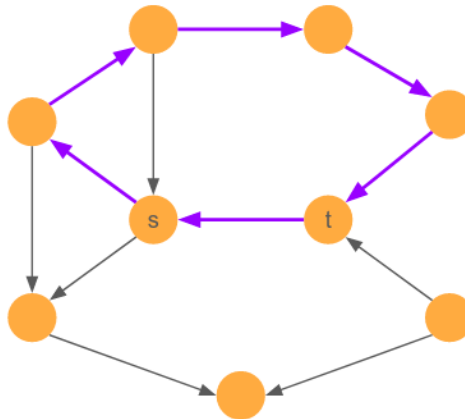


Figure 10: A path from s to t .

We briefly discussed directed graphs, which are more complicated to traverse than undirected ones since we have to respect the direction of the arrows. The road network is also directed, in particular there are **one-way** streets i.e. links from a to b where there is no link from b to a . These one way streets force us to take **detours**, i.e. to get from s to t in Figure 10 we can't go directly, we have to take some other path.

If d_{ij} is the shortest path between i and j on the undirected path let $\vec{d}_{ij}$ be the shortest path on the directed graph. Note that in general $\vec{d}_{ij} \neq \vec{d}_{ji}$. The **detour index** is

$$\bar{\eta} = \frac{1}{N(N-1)} \sum_{i,j} \frac{\vec{d}_{ij}}{d_{ij}} - 1 \quad (5)$$

For an undirected network $\bar{\eta} = 0$. The **detour profile** is

$$\eta(d) = \frac{1}{N(d)} \sum_{i,j \text{ s.t. } d_{ij}=d} \frac{\vec{d}_{ij}}{d_{ij}} - 1 \quad (6)$$

where $N(d)$ is the number of points separated by a distance d .

We can generalise this idea, comparing (shortest) distances on our observed network, which we call **route distances** d_{ij}^R , with some other distances. Since a spatial network has a position vector x_i for every node we can always define $d_{ij}^E = \|x_i - x_j\|$ the **Euclidean distance** which the distance between nodes 'as the crow flies'. We compare this to the route distance via

$$\eta_{ij} = \frac{d_{ij}^R}{d_{ij}^E} \quad (7)$$

Note the detour index above had a -1 , so the minimum value was 0, while the minimum value of the expression above is 1. This is just a convention you have to specify. Either way, we can average this quantity over all neighbours

$$\eta_i = \frac{1}{N-1} \sum_j \eta_{ij} \quad (8)$$

to measure the **accessibility** of node i or over all pairs of nodes

$$\langle \eta \rangle = \frac{1}{N(N-1)} \sum_{i,j} \eta_{ij} \quad (9)$$

as well as a function of distance

$$\eta(d) = \frac{1}{N(d)} \sum_{i,j \text{ s.t. } d_{ij}^E=d} \eta_{ij} \quad (10)$$

A more ‘topological’ approach is to observe that we need at least $N - 1$ edges to connect N nodes in a tree. The **minimum spanning tree**(MST) is the smallest tree that connects all the vertices. If the edges have a weight, e.g. a length d_e , then the MST should minimise

$$L = \sum_{e \in E} d_e \quad (11)$$

we can define the **cost** as

$$C = \frac{L}{L_{MST}} \quad (12)$$

which measures how efficiently the network connects all of its nodes, compared to the minimum possible cost. Adding more links generally makes the network more accessible but also more costly, so there is a trade-off between $\langle \eta \rangle$ and C .

8.4 Edge Orientation

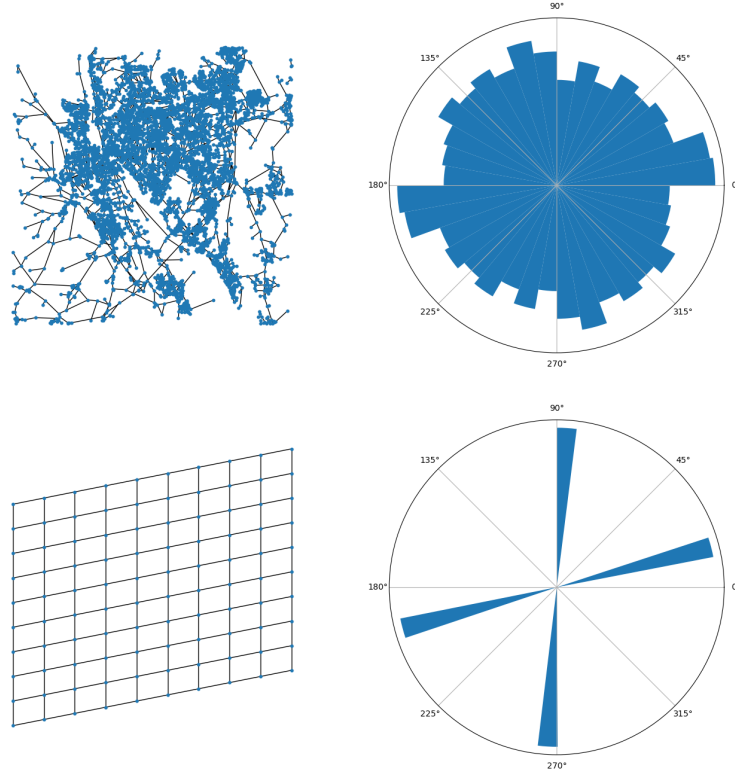


Figure 11: Polar plot for street orientations in Exeter and a regular lattice.

A very nice way to distinguish, say, the regular grid pattern of American streets from the more chaotic street network of European cities is the **edge orientation distribution**. For every link we compute its angle, e.g. for an edge between i and j we have the vector

$$x_{ij} = x_j - x_i = r(\cos \theta, \sin \theta)$$

Plotting a polar histogram of the θ values gives a quite distinct shape for a grid city versus a non-grid one. Figure 11 shows Exeter versus a regular

grid. The histogram for Exeter shows a small excess of roads going east and west, compared to north and south, but is fairly ‘flat’ with roads going in every direction.

8.5 Simple Paths

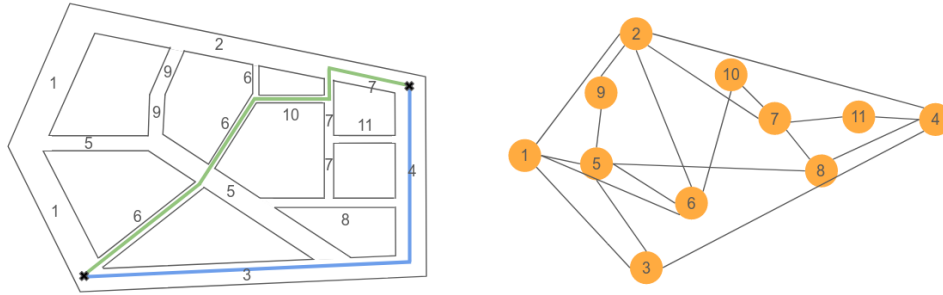


Figure 12: A fictional road network and its dual.

The **simplest path** in a network is the one with the fewest turns. If you look up ‘simple path’ you might come across a different definition, where it means a path without repeated vertices. We will always mean the concept associated with turns in a spatial network. To compute this kind of path with minimal turns we use the dual graph. Recall the dual graph is one where the nodes are streets and they are connected if the streets intersect. Figure 12 shows an example of a short path (green) and a simple path (blue) between two points.

In practice streets are curved, or change names after an intersection so there might be ambiguity in how to do this. The following steps produce a dual network

1. For an edge ij (which will be a node in the dual) we examine its neighbours jk and compute their angles θ_{ij} and θ_{jk} .
2. We find the two edges which are most aligned and compute the angle between them θ_{ijk} .
3. If $\theta_{ijk} < \theta_c$ then the edges ij and jk are assumed to belong to the same street. θ_c has to be chosen and something like 30° is reasonable.
4. We then do the same for the neighbours of i .
5. Once assigned to a street an edge is removed from the network.

To make it unambiguous we start with the longest street, then the second longest and so on until the edges are exhausted.

Define τ as the **number of turns** along a path from s to t which is obtained by counting the the number of switches from one straight line to another on the path. The simplest path therefore is the one with smallest τ which is just the shortest path on the dual graph! If d_{ij} is the length of the shortest path between i and j and d_{ij}^* is the length of the simplest then define the **simplicity index**

$$S = \frac{1}{N(N-1)} \sum_{ij} \frac{d_{ij}^*}{d_{ij}} \quad (13)$$

We can also define the **simplicity profile** as with the detour profile.

$$S = \frac{1}{N(d)} \sum_{ij \text{ s.t. } d_{ij}^E < d} \frac{d_{ij}^*}{d_{ij}} \quad (14)$$

Large values of S mean that the simplest paths tend to be much longer than the shortest ones and the network is hard to navigate. Networks which have high S require a lot of information to specify paths compared to low S ones.

8.6 Spatial Small Worlds

We discussed various models for constructing random networks with different properties, such as ER, WS and BA. These models are mainly focussed on properties related to degree, for example clustering in the WS model or degree distribution in the BA model. There is however a version of the WS model which is of interest for understanding spatial networks. Instead of a ring consider a lattice where all nodes are connected when they are less than some distance p apart. We then add **shortcut** links with probability

$$P_{uv} \sim (d_{uv}^E)^{-\alpha}$$

Where α is some parameter.

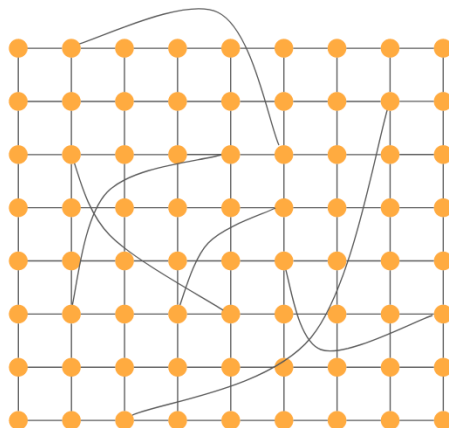


Figure 13: Kleinberg model illustration.

This is called the **Kleinberg model**, see Figure 13. The basic question to ask is, if we are in the network and we have only local information, i.e. we only know the location of our neighbours, how quickly can we pass a message to a target whose position x_t is known? A network where short paths can be found through local searching is called **navigable**.

Kleinberg proposed a greedy search process to pass messages on such a network

1. A message is to be sent to node t whose position x_t is known.
2. A node i which has the message should pass it to the neighbour j which is geographically closest to t .

This is about the simplest algorithm you could think of. We analyse it by studying how the average delivery time $\bar{T}$ varies as a function of N . Through a difficult analytical calculation, and a much easier simulation approach, Kleinberg showed that the delivery time is smallest for $\alpha = 2$. If the exponent is larger, shortcuts are too rare, if it is smaller they are very common and tend to make us hop all over the place. At $\alpha = 2$ the links explore all spatial scales in the network, allowing both long jumps and short hops.

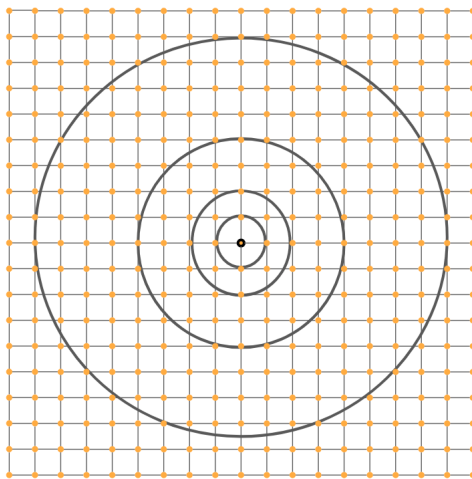


Figure 14: Number of nodes in a ring.

Figure 14 gives some more intuition why 2 is special. We consider a series of ‘zoom levels’, looking at all nodes a distance $1, 2, 4, 8, \dots$ away. Within each newly added ring the number of nodes is proportional to d^2 and the probability of linking to that ring is $d^{-\alpha}$. The probability of a link from the target to any ring is constant only for $\alpha = 2$. For smaller alpha we have more links to further away rings, for larger alpha we preferentially link to closer rings, either of which makes the network more difficult to traverse on average.

8.7 Community Detection

Community detection in spatial networks generally finds communities determined by geography, since most of the flows between nodes take place within the same city, region or country. However community detection can still be useful to *determine* geographic boundaries or to contrast spatial organization with purely topological organization e.g. think of a social network constructed online versus a spatially embedded one. It is much easier for you to shake hands with your friends down the road compared to your friends in Brazil, but it costs about the same to send a message. Thus, the spread of information and the spread of disease might follow quite different trajectories and the communities in a network might be quite different if long links have a large cost.

Recall we discussed modularity in a previous Section

$$Q = \frac{1}{2L} \sum_{ij} \left(A_{ij} - \frac{k_i k_j}{2L} \right) \delta(c(i), c(j))$$

We use this to evaluate the quality of some labelling function c . The term $k_i k_j / 2L$ comes from the configuration model and the modularity counts the excess or deficit of links in a community compared to what would be found in a random graph sampled from the configuration model given with the same labels. We can generalise the modularity to

$$Q = \frac{1}{2L} \sum_{ij} (A_{ij} - P_{ij}) \delta(c(i), c(j))$$

where P_{ij} is the probability of a link between i and j in some null model.

A common model of influence in spatial data analysis is the **gravity model**. Given some measure of importance for nodes I_i (usually population) the influence of i on j is

$$P_{ij}^G = \frac{I_i I_j}{f(d_{ij})} \quad (15)$$

where f is some function which increases with distance and G is a constant. This means the influence of a large population is felt far away and influence between nearby nodes is stronger than between widely separated nodes. Common choices for f are

$$f(d) = d^\alpha$$

or

$$f(d) = e^d$$

If we demand the same number of connections in the observed network and the model at distance d then we can determine it from the data

$$\begin{aligned} \sum_{ij \text{ s.t. } d_{ij}=d} A_{ij} &= \sum_{ij \text{ s.t. } d_{ij}=d} P_{ij}^G \\ &= \sum_{ij \text{ s.t. } d_{ij}=d} \frac{I_i I_j}{f(d)} \\ \rightarrow f(d) &= \frac{\sum_{ij \text{ s.t. } d_{ij}=d} A_{ij}}{\sum_{ij \text{ s.t. } d_{ij}=d} I_i I_j} \end{aligned}$$

i.e. we sum up all the links between nodes separated by distance d and divide by the product of the influences. Thus

$$P_{ij}^G = I_i I_j \frac{\sum_{ij \text{ s.t. } d_{ij}=d} A_{ij}}{\sum_{ij \text{ s.t. } d_{ij}=d} I_i I_j} \quad (16)$$

and

$$Q^G = \sum_{ij} (A_{ij} - P_{ij}^G) \delta(c(i), c(j)) \quad (17)$$

Note, if we have a weighted network (as we usually assume with spatial networks) then A_{ij} are the edge costs/weights.

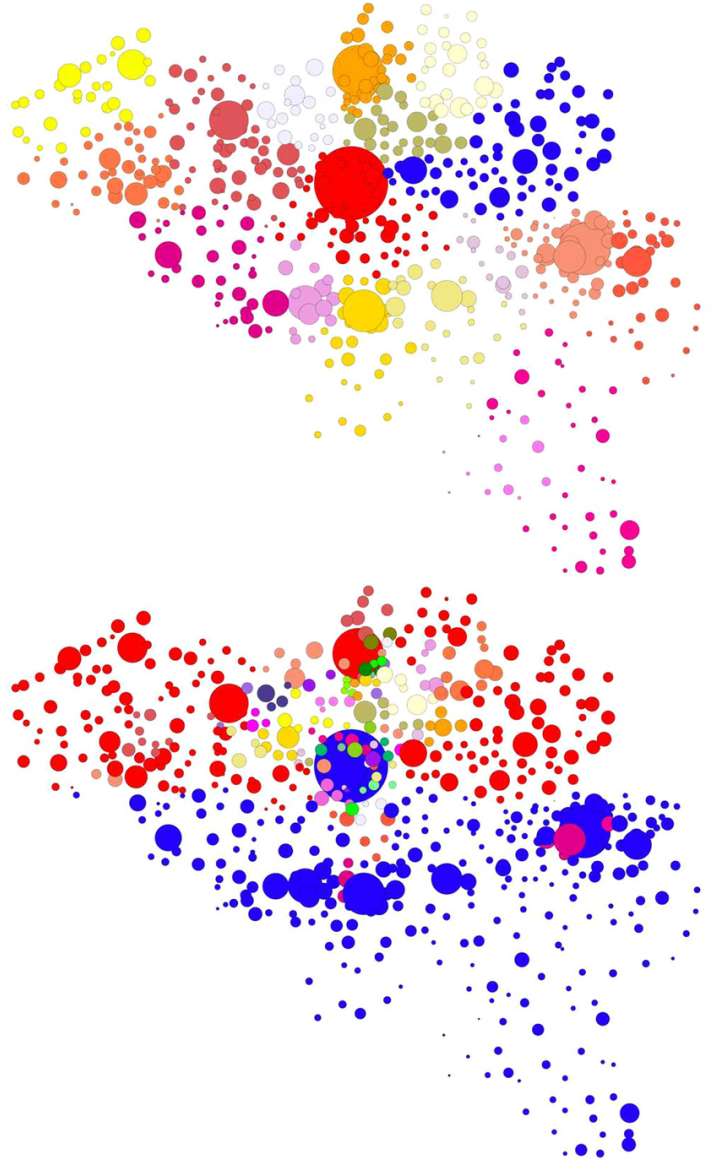


Figure 15: Communities found by optimising Q , top, and Q^G , bottom from [2].

The above was applied to Belgian mobile phone data in [2] where A_{ij} is the number of calls made between various *communes* over a 6 month period. The top, which ignores spatial information, finds several bigger communities

which are spatially coherent i.e. all things considered, you are more likely to call someone who lives nearby. The bottom, which incorporates spatial information finds a different partition, splitting the nodes in two classes corresponding quite closely to the Flemish (north) and French (south) speaking parts of Belgium. The largest city, Brussels, is assigned to the French speaking part. We can also see many other nodes which have been assigned to single node communities. These are places with a lot of mixture between French and Flemish speaking people, and so, with the requirement of a hard partition, they don't fit neatly into red or blue and modularity is improved by putting them in their own communities.

References

- [1] Marc Barthelemy. *Spatial networks: a complete introduction: from graph theory and statistical physics to real-world applications*. Springer Nature, 2022.
- [2] Paul Expert, Tim S Evans, Vincent D Blondel, and Renaud Lambiotte. Uncovering space-independent communities in spatial networks. *Proceedings of the National Academy of Sciences*, 108(19):7663–7668, 2011.